

CAGLIARI ELMAS, Italy

WMO: 165600

Lat: 39.244N

Lon: 9.060E

Elev: 7

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.2	37.2	27.4	21.3	42.4	30.4	24.6	43.0	32.8	57.7	28.0	57.3	6.6	320	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.3	93.5	71.2	91.0	71.0	88.0	70.9	78.5	84.8	76.9	83.3	75.4	82.1	12.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	139.8	82.4	75.0	131.4	81.3	73.3	123.9	79.9	41.9	85.6	40.2	83.5	38.8	82.5	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	21.7	19.3	30.4	100.1	2.3	3.0	28.7	102.2	27.4	103.9	26.1	105.6	24.4	107.7		
(5)			29.0	80.7	2.3	2.6	27.3	82.6	25.9	84.2	24.6	85.7	22.9	87.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	50.4	50.6	54.4	59.0	65.8	73.5	78.7	79.0	72.9	67.1	58.5	52.6		
	DBStd	11.14	4.46	4.52	4.39	3.95	4.35	4.67	3.70	3.66	4.15	4.55	5.02	4.73		
	HDD50	136	49	43	12	1	0	0	0	0	0	0	4	27		
	HDD65	2028	453	403	329	182	43	1	0	0	1	30	201	385		
	CDD50	5109	61	60	149	271	490	705	890	898	687	530	260	108		
	CDD65	1522	0	0	0	2	67	256	425	433	238	95	6	0		
	CDH74	12906	0	0	1	16	359	2059	4197	4329	1593	347	5	0		
	CDH80	4590	0	0	0	0	53	642	1727	1787	353	28	0	0		
(14) Wind	WSAvg	9.1	8.3	8.9	9.7	9.9	10.2	10.0	10.5	9.3	8.6	8.1	7.8	7.9		
(15) Precipitation	PrecAvg	16.6	1.8	2.0	1.6	1.5	0.9	0.4	0.1	0.3	1.3	2.1	2.4	2.2		
	PrecMax	24.8	5.4	6.7	6.4	4.7	4.2	2.7	1.1	1.4	3.9	6.6	7.2	6.3		
	PrecMin	9.4	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.3		
	PrecStd	4.2	1.3	1.5	1.3	0.9	0.9	0.6	0.2	0.4	1.1	1.6	1.7	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.7	66.3	71.7	75.6	85.6	94.9	97.9	98.4	90.0	82.8	74.6	66.6		
		MCWB	58.0	56.2	58.0	59.1	65.0	69.9	70.7	69.2	70.4	69.6	64.3	60.1		
	2%	DB	62.3	63.2	68.0	72.3	80.7	89.7	93.5	93.5	86.2	79.9	71.4	64.6		
		MCWB	56.4	55.2	56.3	59.1	64.4	69.7	70.2	72.3	70.1	68.8	63.6	58.9		
	5%	DB	60.5	61.1	65.7	69.9	77.4	86.3	90.1	91.0	84.0	77.3	69.6	62.8		
		MCWB	55.0	54.4	55.8	58.4	63.3	68.4	70.5	72.1	70.0	67.8	62.9	57.9		
	10%	DB	58.8	59.2	62.9	67.9	75.3	83.8	87.9	88.0	82.0	75.4	67.7	61.0		
		MCWB	53.6	53.1	55.1	57.8	62.6	68.0	70.6	72.3	69.7	67.1	61.8	56.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.8	59.0	61.0	63.7	69.8	77.5	80.1	81.8	77.9	75.4	67.5	62.3		
		MCDB	63.0	63.0	66.9	69.8	76.9	85.1	87.0	87.5	81.7	78.3	70.4	65.0		
	2%	WB	57.6	56.9	59.0	61.7	67.5	74.4	77.7	79.2	75.6	72.6	65.6	60.2		
		MCDB	60.9	61.3	64.5	67.9	75.1	81.9	84.4	85.4	80.6	76.5	69.3	63.4		
	5%	WB	55.8	55.5	57.6	60.4	66.1	72.3	76.0	77.6	74.0	70.5	64.1	58.6		
		MCDB	59.2	59.9	62.6	66.7	73.5	80.5	83.0	84.1	79.7	74.9	68.2	62.2		
	10%	WB	54.3	54.0	56.4	59.3	64.7	70.8	74.5	76.1	72.3	68.9	62.4	57.0		
		MCDB	57.9	58.6	61.2	65.4	72.2	79.0	82.0	83.0	78.8	73.5	66.6	60.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	16.2	17.0	17.1	17.8	18.9	19.6	19.3	17.6	16.3	14.9	14.8		
		MCDBR	16.2	17.8	20.0	20.4	21.3	22.0	22.3	22.5	19.1	17.3	16.3	15.5		
		MCWBR	11.7	11.8	11.5	10.5	9.8	9.4	9.4	10.7	9.2	9.5	10.6	11.2		
	5% WB	MCDBR	14.7	16.3	16.7	17.3	18.3	18.7	18.9	19.1	16.1	14.2	14.5	14.0		
		MCWBR	11.6	11.8	11.3	10.6	10.0	10.4	10.9	11.7	10.3	9.4	10.8	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.365	0.405	0.444	0.459	0.433	0.448	0.438	0.435	0.410	0.382	0.353		
	taud		2.442	2.373	2.257	2.161	2.157	2.277	2.241	2.278	2.279	2.317	2.351	2.409		
	Ebn at Noon		259	270	269	265	263	269	264	264	256	249	241	246		
	Edh at Noon		26	32	40	46	48	42	43	41	38	34	29	25		
(44) All-Sky Solar Radiation	RadAvg		665	947	1352	1737	2080	2392	2418	2124	1593	1146	734	600		
	RadStd		60	82	137	123	146	81	78	99	60	80	52	51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.24	-1.64	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.52	N/A	N/A	N/A	-0.90	-1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page